How Much Semantic Architecture Is Necessary?

A Replay-Controlled Evaluation Protocol for Evidence-Constrained Dataset Schema Induction

Anonymous pre-blind technical report

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Abstract

This work studies the semantic component of a system whose product interface is *dataset in, deterministic schema out*. The physical schema remains parser-owned; language models may only propose property-level semantic additions over existing fields and cited evidence. We replace a historically fragmented semantic pipeline with a controlled four-condition experiment: A is deterministic only; B adds at most one dataset-level semantic call; C reuses the exact B response and applies deterministic evidence-identity, field-relevance, and conflict checks; D preserves the legacy per-field/manual-group schedule and non-empty-evidence merge behavior.

The contribution at the present stage is an auditable protocol and validated measurement harness, not a claim that any architecture wins. On nine synthetic development cases, only two exposed dataset-level semantic opportunities and one opportunity was absent from gold. All pairwise architecture comparisons were therefore correctly declared non-comparable. A smaller Qwen3.5 9B backend added no accepted, scorable claim in this set, making the observed B–C difference uninformative rather than a null effect. Adversarial tests established the intended call, replay, evidence, abstention, applicability, unsupported-claim, and AURC invariants.

Backend qualification subsequently exposed a runtime and measurement failure: an 8,192-token Qwen3.6-27B run with thinking effectively enabled was invalidated. After repairing replay and telemetry provenance, disabling thinking, and loading a 16,384-token context, the same Q4_K_M model passed every predeclared operational criterion over eight non-blind calibration datasets and 74 semantic targets. This qualification establishes execution compatibility only. The component contributions of dataset-level reasoning and deterministic verification remain reserved for an independently annotated, frozen blind evaluation. Explicit provenance placeholders are included for that freeze.

Keywords: semantic schema induction; selective prediction; evidence grounding; deterministic verification; replay identity; model capability debt; dataset metadata; controlled ablation.

1 Introduction

Dataset schema induction contains two different epistemic problems. Physical facts such as field names, stored types, shapes, groups, and declared attributes are properties of bytes and format standards. Contextual meaning—whether a column is an identifier, a measurement, a time axis, or a domain quantity—often requires document-level semantic integration. Treating both problems as free-form language generation makes plausible metadata difficult to distinguish from schema truth. Treating both as deterministic parsing leaves genuinely contextual information unused.

The system studied here therefore retains a strict authority boundary. Deterministic extractors own canonical physical structure. A semantic component may propose claims only for existing fields, cite evidence identifiers from the input record, and abstain when support is insufficient. This design is aligned with provenance models that make derivation auditable [15], evidence-backed model revision [5], attributed generation with explicit support and abstention [13], and selective prediction, where lower coverage can be exchanged for lower risk [7].

The historical implementation used per-field calls, manual grouping, retries, and multiple merge rules. Some of this complexity may have been useful when smaller models struggled to integrate repository- or dataset-level context. We call complexity that remains after its compensating model limitation has weakened *Model Capability Debt*. The scientific question is not whether a newer model can be inserted everywhere. It is whether the minimum architecture that preserves the trust boundary can be identified empirically. This is a project-specific operational construct derived from the broader technical-debt framing for ML systems [17]; it is not asserted as an established universal theory. Khan's single-benchmark prompting study provides a motivating example in which restrictive prompting helped one model generation but hurt a more capable one [8]. That preprint does not validate our construct or establish the pattern for schema extraction. A component counts as capability debt only after a controlled deletion or substitution demonstrates that its compensating value is no longer needed.

We ask three component-attribution questions:

RQ1: What does one dataset-level semantic reasoning call add beyond deterministic extraction?

RQ2: Holding the semantic response exactly constant, what does deterministic verification change?

RQ3: Does the minimal verified hybrid preserve trustworthiness relative to the legacy per-field system while using fewer semantic calls?

At this pre-blind stage, none of these questions has a valid effect estimate. The completed work establishes conditions under which future within-system differences can be attributed to predeclared component substitutions, and identifies precisely what evidence is still missing. This does not identify population-wide or mechanistic causal effects.

2 Related Work and Evaluation Position

Table annotation and dataset metadata. Doduo jointly learns column types and inter-column relations from whole-table context [18]. ArcheType decomposes LLM-based column type annotation into context sampling, prompt serialization, model querying, and label remapping, showing why these choices must be frozen and ablated separately [4]. Korini and Bizer study prompt variants, similarity-selected demonstrations, fine-tuning, model choice, and vocabulary transfer for column property annotation [9]. These systems are important comparators for applicable relational tables, but CTA and CPA do not cover parser-owned physical structure, hierarchical files, arrays, or file-level claims. AutoDDG combines profiled table summaries with LLM enrichment for dataset description and retrieval [21]; description quality is adjacent to, but not equivalent to, schema-claim correctness.

Attribution, abstention, and selective evaluation. Evidence-attributed generation motivates cited support and explicit abstention [13, 5]. A citation is not itself proof of truth, so our verifier checks identity, allowed purpose, field relevance, and contradiction without claiming semantic entailment. Selective-classification work motivates risk–coverage reporting [7], while Traub et al. show that common multi-threshold metrics can change system rankings and propose AUGRC as an alternative [20]. We therefore keep exact accepted claim rate as the primary endpoint, report AURC only with achieved coverage and tie rules, and reserve AUGRC as a predeclared descriptive sensitivity.

Validity, annotation, and resource cost. Benchmark contamination can inflate apparent LLM performance even when exposure is difficult to measure [16]; a sealed split prevents project-side tuning but does not prove absence from model pretraining. Annotation anchoring can alter agreement [1], motivating independent prediction-blind submissions that are frozen before disagreement reveal. Cost-aware LLM routing demonstrates that quality must be interpreted jointly with resource use [2]. We report calls, retries, tokens, latency, configured cost, and effective architecture cost, but do not treat a learned model cascade as a direct baseline for this study.

3 Problem Formulation and Product Boundary

The product-level mapping is

$$\texttt{dataset file(s)} \longrightarrow \texttt{deterministic canonical schema} \oplus \texttt{optional semantic claim ledger}.$$

The deterministic schema is a complete standalone output. Semantic augmentation is an optional overlay and may never invent fields, silently overwrite stronger deterministic claims, or convert missing support into confidence.

For field path f, property p, and proposed value v, a semantic claim is represented as

$$c = (f, p, v, E, s, d),$$

where E is a set of evidence identifiers, s is a confidence value used for risk–coverage ordering, and d is the final decision. Correctness and confidence are deliberately separate: a claim may be confident but unsupported, or supported but uncertain. Decisions include accepted, rejected, and abstained; verification states distinguish verified agreement, evidence-grounded support, contradiction, conflict, and unsupported status.

Gold applicability is likewise explicit. Each field–property slot is one of: (i) applicable with a value, (ii) applicable but legitimately unknown, (iii) not applicable, or (iv) outside an open-world gold scope. This prevents an abstention from being silently scored as an incorrect accepted value and prevents missing gold coverage from being misread as N/A.

4 First-Principles Architecture Audit

The audit separated essential from accidental complexity. Parser authority, structured abstention, property-level provenance, and visible conflict states are essential because they encode trust boundaries. Per-field prompt fragmentation, manual grouping, repeated free-JSON/schema attempts, and redundant transformations are historical until their value is demonstrated. Rigid output contracts remain useful where they make failures observable; they are not justified merely as a way to force a preferred answer.

The audit produced one conservative redesign rule: delete semantic orchestration only after a replay-controlled experiment shows that a simpler approach is equal or better. This avoids replacing old Model Capability Debt with new model dependence.

5 The Four Fixed Architecture Variants

Table 1: Frozen semantic architecture variants. No additional variant is permitted in the study.

ID	Name	Semantics	Semantic generation calls
A	Deterministic only	Emits parser-owned schema and deterministic claims; no semantic model.	0
B	Dataset reasoner	Deterministic extraction plus at most one dataset-level semantic response for all unresolved targets.	≤ 1 per dataset
C	Replayed reasoner + verifier	Consumes the exact B response; checks evidence identity, field relevance, deterministic contradiction, and cross-claim conflict.	0 new calls
D	Legacy semantic pipeline	Preserves per-field/manual-group scheduling, temperature 0.2, free-JSON then schema fallback, and the historical non-empty-evidence merge rule.	Variable, historically many

The B–C contrast is the cleanest intervention. A memoizing backend keys the full request and provides C with the exact raw response and SHA-256 identity produced for B. C is invalid if it generates a response, lacks a replay, or consumes a different hash. Verification is deterministic and does not claim semantic entailment: an existing, field-relevant citation establishes traceable support, not factual truth.

6 Evaluation Methodology

Let V be applicable-value slots, P accepted predictions on those slots, and $K \subseteq P$ correct accepted predictions. We report

$$\text{end-to-end accuracy} = \frac{|K|}{|V|}, \tag{1}$$

$$\text{coverage} = \frac{|P|}{|V|}, \tag{2}$$

$$\text{selective risk} = 1 - \frac{|K|}{|P|}. \tag{3}$$

When P is empty, selective accuracy/risk is undefined rather than forced to zero. Abstentions lower end-to-end accuracy and coverage but do not enter the conditional selective-risk denominator. Confidence is used only to order whole tie groups for a right-continuous risk–coverage curve. AURC is integrated only over achieved coverage. In the NDP-50 path, score-v3 binds every accepted applicable-known slot to its arm-specific confidence and exact-canonical correctness. Fixed-decile reliability bins are reported separately by arm and property; Brier score, fixed-bin ECE, and calibration wording are permitted only at 30 or more accepted observations in that group. Cross-arm and cross-property pooling is forbidden. Low-support groups retain their support, bins, achieved coverage, curve, and descriptive AURC but cannot support a calibration claim. Confidence remains an ordering or probability claim according to its frozen source declaration, never evidence or verification.

Unsupported-claim rates are computed over model-proposed or model-accepted claims, never diluted by deterministic claims. Cost accounting separates generated calls, reused upstream calls, and effective architecture cost. A pairwise comparison is non-comparable if either run is partial, replay identity fails, required gold is missing, or a semantic target lies outside the scoring design.

6.1 Adversarial evaluator validation

The focused suites now contain 53 tests across the variant and backend qualification modules, within 527 passing project tests. They cover nonexistent, irrelevant, wrong-span, contradictory, and empty evidence; confidence 0 and 1; malformed and partial output; duplicates and conflicting claims; applicable unknown and N/A; missing or mismatched replay; hidden retries; timeout; zero/all accepted predictions; and confidence ties. Additional regressions protect failure replay identity, per-call context accounting, legacy fallback telemetry, and partial D provenance.

Table 2: Core component-attribution invariants enforced by code and tests.

Invariant	Enforced behavior
A call boundary	Always zero model calls.
B call boundary	At most one dataset-level generation call; hidden retries make the run partial.
C replay boundary	Zero generation calls and exact B raw-response hash, including malformed-response replay.
Evidence boundary	Missing, empty, cross-field, or wrong-span evidence cannot verify a claim.
Confidence boundary	Confidence never implies verification.
Applicability boundary	N/A, applicable unknown, applicable value, and outside-scope are distinct.
Comparability boundary	Partial execution, missing gold coverage, or replay failure suppresses component-attribution deltas.
Legacy fidelity	Historical scheduling and merge behavior remain unchanged; all attempts are measured.

7 Synthetic Development Experiment

The development manifest contains 9 authored regression cases spanning CSV, HDF5, and time-series structures. It is not a random sample and provides no external-validity evidence. Only 2/9 cases contain any dataset-level semantic target. One of those cases targets `/campaign/meta/platform`, which is absent from gold; consequently every formal A–B, B–C, and C–D comparison is non-comparable.

The local development backend was Qwen3.5 9B. The two actual B responses were replayed byte-for-byte into C; the other seven cases correctly made no B/C call. Table 3 reports pooled descriptive diagnostics, not architecture effects.

Table 3: Nine-case synthetic development diagnostics. All B/C/D component comparisons are non-comparable.

Variant	Comparable	End-to-end acc.	Coverage	Sel. risk	AURC	Calls
A	yes	0.976608	0.988304	0.011834	0.014603	0
B	no	0.976608	0.988304	0.011834	0.014603	2
C	no	0.976608	0.988304	0.011834	0.014603	0 new / 2 effective
D	no	0.988304	1.000000	0.011696	0.014705	6

B proposed one model claim, accepted none, and added no scorable coverage. Its effective cost was 5752 input tokens, 381 output tokens, and 4128.415 ms of model latency. C had the same effective upstream cost and zero new generation. D proposed and accepted four claims, used 6 calls, 7538/3598 input/output tokens, and 22171.453 ms. D recovered two gold-covered logical values in one hard CSV case, but another change was unscored. These facts do not establish superiority because the design is non-comparable.

The Qwen3.5 backend abstained on a postal-code target even though its notes recognized the semantics, and violated target scope on the unscored HDF5 case. A stronger model might resolve either failure without architecture changes; this is a hypothesis, not a result. The development run therefore yields no estimate of the increment from A to B and no verification trade-off from B to C. The all-zero observed B–C decision delta is uninformative, not evidence of no verifier effect.

8 Backend Qualification

Qualification asks whether a backend can execute the frozen contracts, not whether its claims are correct. The fixed non-blind calibration manifest contains 8 external datasets and 74 semantic targets. It is excluded from the future blind benchmark. B is run twice per dataset at temperature 0 and seed 0; C immediately replays each response; D runs once with its preserved policy.

8.1 Invalidated 8K/thinking-on execution

The first Qwen3.6-27B Q4_K_M execution used an 8,192-token loaded context while thinking remained effectively enabled. Only 2/16 B attempts and 1/8 D dataset runs were contract-valid; 12 B attempts terminated at the generation boundary, and a 10,044-token HDF5 prompt was rejected before generation. The run also exposed three measurement defects: malformed B failures did not carry raw identity into C, unparseable output could be counted target-scope valid, and partial D failures dropped earlier-group/fallback provenance. The immutable artifact was invalidated; none of its rates is used for backend selection.

The failure was informative as infrastructure diagnosis. LM Studio 0.4.18 documents a fix for reasoning on/off being ignored by some custom configurations [11]. We repaired only provenance and measurement, not prompts, architectures, targets, or call budgets.

8.2 Corrected 16K/thinking-off execution

A one-call gate first established a contract-valid response with zero reasoning tokens. The same Qwen3.6-27B Q4_K_M model [12] was then run with thinking disabled and a 16384-token context. It passed every frozen criterion (Table 4).

Table 4: Corrected Qwen3.6-27B operational qualification.

Criterion	Observed	Required	Result
Semantic-opportunity datasets	8	≥ 8	pass
B contract-valid rate	1.0	≥ 0.95	pass
B target-scope-valid rate	1.0	1.0	pass
B/C replay identity	1.0	1.0	pass
B exact repeatability	1.0	1.0	pass
B timeout / truncation rates	0.0 / 0.0	0.0 / 0.0	pass
B context fit / observed telemetry	1.0 / 1.0	1.0 / 1.0	pass
D contract-valid rate	1.0	≥ 0.95	pass
D timeout / truncation rates	0.0 / 0.0	0.0 / 0.0	pass
D context fit / observed telemetry	1.0 / 1.0	1.0 / 1.0	pass

All 16 B attempts ended with JSON and `finish_reason=stop`; repeats were byte-identical for all eight datasets. Every C input hash matched B and every C semantic generation count was zero. The maximum observed B context was 13306 tokens; the maximum requested `input+max_tokens` budget was 14142, below the loaded limit.

Table 5: Descriptive operational cost in the corrected qualification.

Condition	Calls	Input tokens	Output tokens	Mean latency
B (two repeats)	16	76976	26436	27.95 s per attempt
C generation	0	0 new	0 new	deterministic replay/verification
D (one run)	59	79571	19547	44.61 s per dataset

D's 59 calls are an observed consequence of historical scheduling: the HDF5 case alone required 22 field calls. One `functional_traits.csv` free-JSON shape attempt failed and the preserved schema fallback succeeded. The cost gap is real; an accuracy–cost trade-off has not yet been estimated.

9 What the Current Results Do and Do Not Establish

Established facts. The harness enforces the four fixed architectures, exact B/C replay, zero C generation, fail-closed contracts, explicit applicability, and per-call telemetry. The Qwen3.6-27B Q4_K_M condition is operationally eligible under the corrected runtime. The legacy architecture has a substantially larger call surface on the calibration tasks.

Not established. The study has no blind effect estimate. The nine-case development set is too sparse and incompletely gold-covered to estimate A–B, B–C, or C–D effects. Backend qualification contains no gold. It cannot show that dataset-level reasoning helps, that verification lowers risk, that D is more accurate, or that Qwen generalizes.

Model selection. There is no operational reason to replace Qwen3.6-27B with GLM, Llama, or a smaller Qwen. Alternative families may be predeclared sensitivity conditions, but adding them post hoc to rescue an unfavorable architecture result would confound backend and architecture selection. Model cards and exact runtime identity should be reported as first-class artifacts [14].

10 Blind External Evaluation Protocol

Two qualified annotators who did not develop or actively review the system will independently label the blind gold. They will distinguish applicable value, applicable unknown, N/A, and outside-scope slots. Both submissions are content-hashed before a deterministic disagreement report is revealed; consensus then resolves every disagreement without access to architecture outputs. Pre-consensus reporting includes exact agreement and support by decision slot, disagreement counts, and Cohen's kappa with marginal tables [3]. Degenerate kappa is reported as undefined, not zero. Krippendorff's alpha [10] may be added descriptively only after its unitization, distance function, and missing-data policy are frozen; no agreement coefficient replaces reconciliation.

Dataset-level paired effects, not pooled slots alone, will be the primary analysis unit. The development set will not be used as evidence of generalization. Dataset documentation follows the broader principle that collection, intended use, and limitations should be explicit [6].

The primary contrasts are predeclared:

- A–B: added correct/incorrect claims, coverage, selective risk, unsupported claims, successful/failed calls, retries, tokens, model latency, end-to-end latency, and cost;
- B–C: B-accepted claims rejected or abstained by C, incorrect claims removed, correct claims lost, unsupported claims removed, and the risk–coverage tradeoff;
- C–D: quality, selective metrics, unsupported claims, successful/failed calls, retries, tokens, both latency measures, cost, and failure surface, only when D is comparable.

The deterministic condition is the primary non-LLM baseline. Human consensus is the reference standard, not a human upper bound. Doduo and ArcheType are scoped literature comparators for applicable table-annotation cases, not drop-in baselines for heterogeneous file extraction. An evidence-free model call, a name/value heuristic, or a second model family may be useful as a secondary sensitivity, but none may be added after outcomes or used to rescue the primary result. Each requires a versioned design, fixed information boundary, complete paired execution, and updated multiplicity/power review before activation.

10.1 NDP-50 operationalization

The external protocol is instantiated on a content-addressed National Data Platform candidate frame of 5,823 catalog records from 75 organizations. The frozen sample contains 50 datasets: 15 development, 10 validation, and 25 sealed test. Selection uses format strata, an organization cap, and SHA-256 priorities. The dataset is the statistical unit; multiple resources remain clustered within their source dataset. At this manuscript revision, structural validation is complete within its declared scope, while semantic execution and test opening are not authorized.

The NDP-50 protocol retains two co-primary contrasts: deterministic-only versus dataset-level zero-shot augmentation, and zero-shot versus deterministic verification of its byte-identical response. Few-shot prompts, row samplers, and transfer analyses remain descriptive. A public OSF or Zenodo registration receipt must bind the final public protocol, analysis plan, and artifact hashes before test release, without exposing sealed test identities.

The local registration-source builder hashes a sorted text allowlist and rejects path escape, known test-detail locations, and exact matches to sealed dataset IDs or titles. This is a disclosure-control check, not an independent timestamp and not proof against indirect re-identification; the uploaded attachment set still requires human review and an immutable external receipt.

The registered NDP-50 primary p-values use paired two-sided sign-flip tests on dataset-level differences with Holm control across the two contrasts. This tests sign exchangeability about zero; it is not distribution-free for a mean-null under arbitrary asymmetry. We therefore precommit to reporting the full paired-effect distribution, the mean effect and dataset-bootstrap interval, ties, noncomparability, and assumption diagnostics. The equal-weight estimand targets the quota-weighted NDP-50 design, not prevalence in the complete catalog. A failure to reject will not be described as evidence of no effect.

Resource comparisons are also preregistered rather than reconstructed post hoc. The execution freeze binds the pricing basis, effective date and source, exact call/token rates, treatment of failed, retried, and replayed calls, and the selected backend's hardware/runtime identity. Score records separate model latency from end-to-end latency. When local compute is frozen as not monetized, USD cost is reported as zero with that explicit boundary while calls, tokens, failures, retries, latency, and hardware remain visible; this is not interpreted as zero resource use.

The two semantic-gold annotators must pass a preregistered non-NDP calibration round bound to the final handbook and NDP vocabulary before seeing blind NDP-50 packets. Their passing summary, external pre-submission receipt, identities, and independence attestations are release-gate artifacts; agreement coefficients remain process evidence rather than accuracy.

10.2 Blind-freeze provenance gate

STOP: blind gold and test release remain blocked.
External preregistration, calibrated independent annotation, adjudicated gold, backend/checkpoint identity, source/analysis freezes, and independent countersignatures are unresolved. Every item in the full binding ledger (Appendix B) must be content-addressed and verified; no development or qualification artifact may silently substitute for it.

The external-registration requirement is machine-enforced rather than prose-only. The readiness and test-release validators require both a non-independent collaborator sign-off covering all 16 feedback decisions and an immutable OSF/Zenodo receipt for the exact public-package digest, separately checked by a non-developer who is not a project collaborator. Neither artifact alone authorizes test opening, and both are currently absent. The content-bound collaborator assignment is released for review, but its neutral payload contains no signature or human decision. Collaborator sign-off, external registration, and test authorization are separate ordered stages so that no required pretest artifact is circularly locked behind `test_ready`.

The known pre-freeze contract hashes are reported in Appendix A; they do not substitute for the final registry, checkpoint, source-tree, or gold receipts.

11 Threats to Validity

Construct validity. Evidence identity and field relevance do not prove semantic entailment. C should be interpreted as deterministic trust-boundary verification, not an oracle. Confidence values may be incomparable across properties or arms; the NDP contract therefore prohibits pooling and suppresses Brier/ECE calibration claims below the frozen 30-observation arm-by-property threshold. AURC is secondary, is integrated only over achieved coverage, and may rank systems differently from AUGRC or the primary endpoint.

Internal validity. B/C replay is tightly controlled, but D intentionally retains temperature 0.2 and historical fallback behavior. Backend state, runtime version, context, checkpoint, and prompt hashes must remain frozen. Any partial response or missing replay suppresses the corresponding comparison.

External validity. The nine development cases are synthetic and the eight qualification cases are a fixed operational calibration set already exposed to development. Neither is a population sample. Only the future independently annotated corpus can support bounded generalization claims. Test sealing prevents project-side adaptation but cannot establish that a pretrained model never encountered related public data.

Annotation validity. Independent, prediction-blind submissions reduce anchoring and collaboration conflicts but do not guarantee correct labels. Agreement, disagreement, evidence identity, reconciliation rate, and unresolved scope must all be reported. Active collaborators and developers cannot be counted as independent annotators, fresh adjudicators, independent methods reviewers, or test-release monitors.

Statistical conclusion validity. The blind analysis must use paired dataset-level effects and confidence intervals. A clean null result is acceptable. The current manuscript reports descriptive pooled metrics only to document harness behavior.

Reproducibility. The exact checkpoint-file hash and machine-observed LM Studio version are still missing. The version is operator-reported. These are explicit freeze blockers rather than hidden caveats.

12 Conclusion

The durable system objective remains dataset input to deterministic schema output, with optional evidence-constrained semantic enrichment. The present work reduces the semantic architecture question to four auditable variants and validates the measurement boundaries needed for controlled component attribution. Development data do not rank the variants. Corrected backend qualification shows that one dataset-level Qwen3.6-27B response can be generated repeatably and replayed into deterministic verification, while the legacy condition exposes a much larger call surface.

The central research answer is intentionally deferred: whether semantic reasoning adds useful claims, and whether deterministic verification removes more errors than correct claims, can only be learned after the blind provenance block is completed and independent gold is unsealed. Until then, the scientifically correct result is a validated protocol, an operationally eligible backend, and no architecture winner.

A Content-Addressed Current Artifacts

Artifact	SHA-256
Development manifest	a1a199272b9019dd99dab873239f18ba1d8b656235c54ac40ad84fb5d1738797
Development report	1e77f66d924622153798743c209497b5025271c5d94ad4cebf6290871ff87e27
Qualification manifest	71aa69e25fddf8a3cfe9ad78e2cc2ba08015347f8ea53166985c081120fe2666
Controlled vocabulary	874548a10f8871449be9b749c99e95e72e8852d261324abc8fa3c492246ed888
Invalidated 8K qualification	d0a5fb9a82d41bda557669fd384820485c5265f187679f2e954ad347eec25cde
Corrected qualification	c89145cf23943fd2991504a31cea5a10ceddccc90725e31932cc02770b1c0538
Qualification assessment	bec8279af416b796d4ea5fe071a8947fcd1d7d123c1cc021168dd928fcb87efb
Dataset reasoner prompt	1b75583a35549a5cbdfecc79be5796eb358d2e5f976f5483b25756a81f8f8060
Dataset response schema	e79212874554daf93c401578b97963f45934854b22b6786ee41ae60bf8dfd7ca
Legacy prompt	3d4047f7bdc36e6ae832fe7e988bd06bff39be81f73bea101c6abdfe32b78a0f
Legacy response schema	83ee4fb51579af04a304c61eb4b610263b1d02fcae1a90a56cfdb131e445a91f

B Full Blind-Freeze Binding Ledger

Required binding	Current value
Blind sampling registration receipt	<TO BE BOUND BEFORE GOLD UNSEALING>
External preregistration DOI/receipt and timestamp	<TO BE BOUND BEFORE GOLD UNSEALING>
Blind candidate-frame SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Blind manifest SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Independent annotation package SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Adjudicated gold package SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Annotation vocabulary/handbook SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Annotator pseudonyms and independence attestations	<TO BE BOUND BEFORE GOLD UNSEALING>
Calibration-gate result and agreement statistics	<TO BE BOUND BEFORE GOLD UNSEALING>
Frozen backend-registry SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Exact Qwen GGUF checkpoint SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
LM Studio 0.4.18 runtime/version receipt SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>
Frozen source commit/tree SHA-256	<TO BE BOUND BEFORE GOLD UNSEALING>

Required binding	Current value
Frozen analysis-plan SHA-256	`<TO BE BOUND BEFORE GOLD UNSEALING>`
Prompt/schema bundle SHA-256	`<TO BE BOUND BEFORE GOLD UNSEALING>`
Gold unsealing timestamp and countersignatures	`<TO BE BOUND BEFORE GOLD UNSEALING>`

C Current Reproduction Status

At manuscript generation, 527 project tests passed. The relevant Ruff checks passed, all 316 JSON artifacts parsed successfully, and `git diff --check` passed. Seven pre-existing full-tree lint findings in unrelated legacy files remain documented and were not modified. The two PDFs were rendered from these TeX sources using Tectonic 0.16.9 [19].

The feedback response matrix, claim ledger, and OSF/Zenodo preregistration draft are publication-control artifacts. The registration document remains explicitly unsubmitted: a local draft or placeholder is not evidence of preregistration and adds no test-release authority.

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